On Generative Agents in Recommendation

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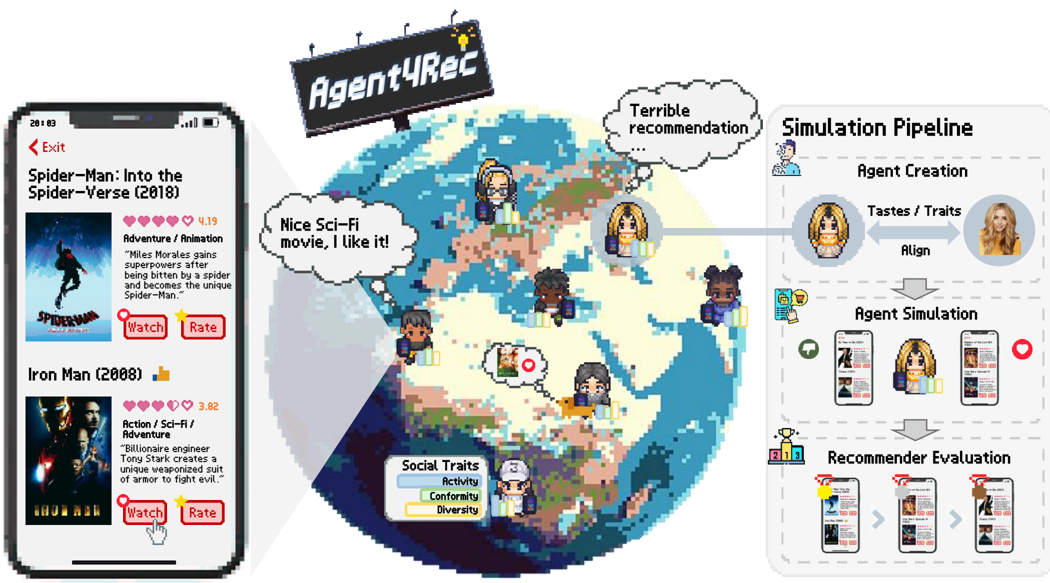

Figure 1: Illustration of Agent4Rec, a user simulator with 1,000 LLM-empowered generative agents in the movie recommendation scenario. These agents are initialized from the MovieLens-1M [15] dataset, embodying varied social traits and preferences. Each agent interacts with personalized movie recommendations in a page-by-page manner and undertakes various actions such as watching, rating, evaluating, exiting, and interviewing. With Agent4Rec, we would like to explore the potential of LLM-empowered generative agents in simulating the behavior of genuine, independent humans in recommendation environments.

ABSTRACT

Recommender systems are the cornerstone of today's information dissemination, yet a disconnect between offline metrics and online performance greatly hinders their development. Addressing this challenge, we envision a recommendation simulator, capitalizing on recent breakthroughs in human-level intelligence exhibited by Large Language Models (LLMs). We propose **Agent4Rec**, a user simulator in recommendation, leveraging LLM-empowered generative agents equipped with user profile, memory, and actions modules specifically tailored for the recommender system. In particular, these agents' profile modules are initialized using real-world datasets (*e.g.,* MovieLens, Steam, Amazon-Book), capturing users' unique tastes and social traits; memory modules log both factual and emotional memories and are integrated with an emotion-driven reflection mechanism; action modules support a wide variety of behaviors, spanning both taste-driven and emotion-driven actions. Each agent interacts with personalized recommender models in a page-by-page manner, relying on a pre-implemented collaborative filtering-based recommendation algorithm. We delve into both the capabilities and limitations of Agent4Rec, aiming to explore an essential research question: "To what extent can LLM-empowered generative agents faithfully simulate the behavior of real, autonomous

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关于推荐中的生成式智能体

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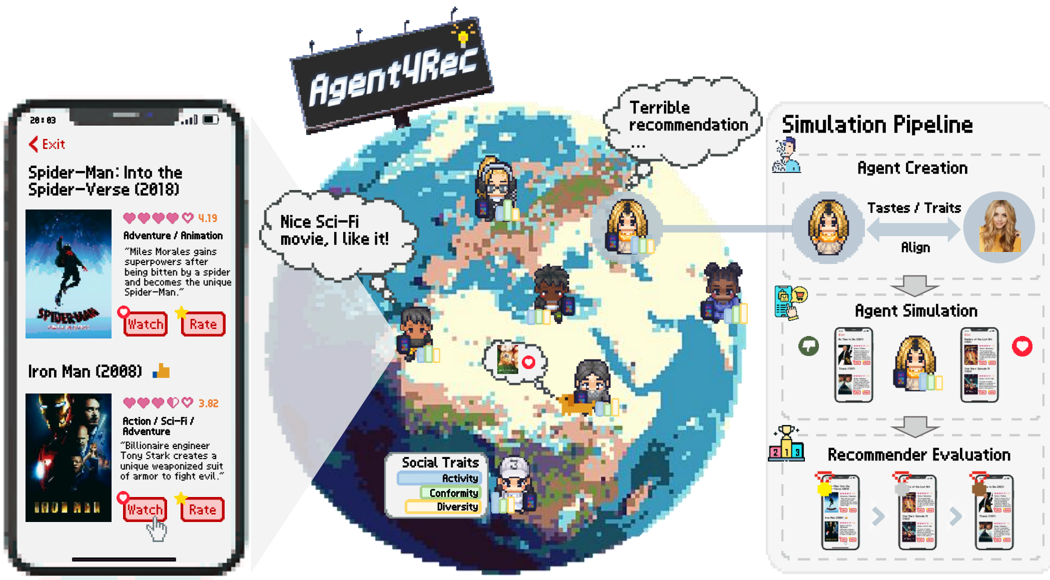

图1：Agent4Rec 示意图——一个在电影推荐场景中由 1,000 个由 LLM 赋能的生成式智能体组成的用户模拟器。这些智能体基于 MovieLens-1M [15] 数据集进行初始化，体现出多样的社会特征与偏好。每个智能体以逐页的方式与个性化电影推荐进行交互，并执行观看、评分、评估、退出、访谈等多种动作。借助 Agent4Rec，我们希望探索由 LLM 赋能的生成式智能体在推荐环境中模拟真实、独立人类行为的潜力。

摘要

推荐系统是当今信息传播的基石，然而离线指标与在线表现之间的脱节极大地阻碍了其发展。为应对这一挑战，我们设想构建一个推荐模拟器，利用近期在人类水平智能方面所展现的突破性进展。

大型语言模型（LLMs）。我们提出 Agent4Rec，一种用于推荐场景的用户模拟器，利用由 LLM 赋能的生成式智能体，并配备专为推荐系统量身定制的用户画像、记忆与行动模块。具体而言，这些智能体的画像模块使用真实世界数据集（例如 MovieLens、Steam、Amazon-Book）进行初始化，以捕捉用户的独特品味与社会特征；记忆模块记录事实性与情感性记忆，并与一种情绪驱动的反思机制相结合；行动模块支持多种多样的行为，涵盖由品味驱动与由情绪驱动的行动。每个智能体以逐页的方式与个性化推荐模型交互，依赖于一个预先实现的基于协同过滤的推荐算法。我们深入探讨 Agent4Rec 的能力与局限，旨在探索一个关键研究问题：$v*$ 在多大程度上，LLM 赋能的生成式智能体能够忠实地模拟真实的、自主的

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humans in recommender systems?" Extensive and multi-faceted evaluations of Agent4Rec highlight both the alignment and deviation between agents and user-personalized preferences. Beyond mere performance comparison, we explore insightful experiments, such as emulating the filter bubble effect and discovering the underlying causal relationships in recommendation tasks. Our codes are available at https://github.com/LehengTHU/Agent4Rec.

CCS CONCEPTS

• **Information systems** → **Recommender systems**.

KEYWORDS

Recommender System, Large Language Model, Generative Agents

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1 INTRODUCTION

Recommender systems play a pivotal role in contemporary information dissemination, actively shaping individual preferences and cognitive processes [29, 63]. Despite their great success and widespread adoption, the conventional supervised recommendation approach falls short, as evidenced by the significant gap between offline metrics and online performance [64, 67]. This disconnect hinders the integration of academic research into real-world recommendation deployments, acting as a bottleneck for the field's future advancements [13, 23]. Imagine a paradigm where a configurable simulation platform for recommender systems exists — one that faithfully captures user intent and encodes human cognitive mechanisms. Such a simulator undoubtedly has the potential to revolutionize traditional research paradigms in recommendations, offering an innovative pathway for data collection, recommender evaluation, and algorithmic development [6, 20, 50].

Recent strides in Large Language Models (LLMs), with their impressive capability [40, 47, 58] and profound comprehension of time and space [14, 53, 56], underscore the promise of the recommendation simulator paradigm. Specifically, LLMs serve as the foundational architecture in the development of generative agents [9, 42, 44]. These agents are then integrated into a recommendation environment, taking on the virtual users in simulators. Yet, in this promising research direction, developing a reliable simulator that faithfully mirrors personalized user preferences is non-trivial [54, 69]. Consequently, harnessing the potential of LLM-empowered generative agents by designing modules tailored for recommendation to emulate human behavior has become a key research focus.

To bridge this gap, we introduce **Agent4Rec** — a general user simulator in recommendation scenarios, which consists of two core facets: LLM-empowered generative agents and recommendation environment (*cf.* Figure 2). From the user's perspective, we simulate 1,000 LLM-empowered generative agents per recommendation scenario, where each agent is initialized based on real-world datasets and composed of three essential modules: the user profile, memory, and action modules. The profile module functions as a repository for personalized social traits and historical preferences [38], facilitating the alignment of user portraits with genuine human characteristics. The memory module records past viewing behaviors, system interactions, and emotional memories (*i.e.,* user feelings and fatigue levels) in natural language, enabling information retrieval, preference accumulation, and emotion-driven reflection in a coherent manner. The action module empowers these agents to interact directly with the recommendation environment, including taste-driven actions (*i.e.,* viewing or ignoring recommended movies, rating, generating post-viewing feelings), and emotion-driven actions (*i.e.,* exiting the system, evaluating recommendation lists, and expressing human-understandable comments). From the perspective of the recommender system simulation, items are recommended by a predetermined recommendation algorithm, sequenced in a page-by-page format to emulate real-world recommendation platforms. On the one hand, our simulator predominantly integrates collaborative filtering-based recommendation strategies, encompassing random, most popular, Matrix Factorization (MF) [24], LightGCN [16], and MultVAE [28]. On the other hand, we architect the simulator with extensibility as a core principle. By providing open interfaces, we empower researchers and practitioners to effortlessly deploy any recommendation algorithm of their choice.

To systematically evaluate the effectiveness and limitations of our proposed Agent4Rec, we conduct comprehensive experiments from both the user's and recommender system's perspectives. From the user's standpoint, our primary focus lies in assessing the degree of agent alignment. Specifically, we evaluate to what extent the agent can ensure the coherence of the true user's social traits, personality, and preferences using a variety of metrics and statistical tests. On the recommender system simulation front, we evaluate various recommenders configured with different algorithms. Our evaluation metrics span multiple dimensions, including the average number of items recommended and watched by users, average user ratings, user engagement time, and overall user satisfaction. In parallel, the agent feedback collected from the simulator serves as augmented data, enabling iterative training and refinement of the recommendation strategies. Subsequently, we assess the feedback-driven recommender enhancement using standardized offline metrics. This dual approach, combining simulation feedback with traditional offline evaluation, ensures a comprehensive assessment of recommendation algorithms.

To explore the potential of the simulator in investigating unresolved challenges in recommendation tasks, we undertake two experiments. In the first experiment, we emulate the filter bubble effect within the simulator — a scenario where users are consistently exposed to similar or reinforcing content, resulting in a reduction in item attribute diversity [10, 27]. This investigation aims to understand the extent to which feedback loops can amplify such centralized recommendation phenomena. Additionally, we utilize the simulator as a data collection tool to pioneer a data-oriented causal discovery [17, 62]. This approach yields a robust causal graph, enabling us to unveil the intricate latent causal relationships that infer the data generation progress in recommender systems [11].

Our main contributions are summarized as follows:

在推荐系统中的人类？”对 Agent4Rec 进行的广泛且多维度评估凸显了智能体与用户个性化偏好之间既一致又偏离的情况。除了单纯的性能对比之外，我们还探索了一些富有洞见的实验，例如模拟过滤气泡效应，以及发现推荐任务中潜在的因果关系。我们的代码可在 https://github.com/LehengTHU/Agent4Rec 获取。

CCS 概念

- 信息系统 → 推荐系统。

关键词

推荐系统、大型语言模型、生成式智能体

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1 引言

推荐系统在当代信息传播中发挥着关键作用，积极塑造个体偏好与认知过程[29, 63]。尽管其取得了巨大成功并被广泛采用，传统的有监督推荐方法仍显不足，这一点可由离线指标与在线表现之间的显著差距所证明[64, 67]。这种脱节阻碍了学术研究与真实世界推荐部署的融合，成为该领域未来发展的瓶颈[13, 23]。设想这样一种范式：存在一个可配置的推荐系统仿真平台——它能够忠实捕捉用户意图，并对人类认知机制进行编码。这样的模拟器无疑有潜力革新传统的推荐研究范式，为数据采集、推荐评估与算法开发提供一条创新路径[6, 20, 50]。

大型语言模型（LLMs）的最新进展，凭借其令人印象深刻的能力[40, 47, 58]以及对时间与空间的深刻理解[14, 53, 56]，凸显了推荐模拟器范式的潜力。具体而言，LLMs 作为生成式智能体开发的基础架构[9, 42, 44]。随后，这些智能体被集成到推荐环境中，在模拟器里扮演虚拟用户。然而，在这一充满前景的研究方向中，开发一个能够忠实映射个性化用户偏好的可靠模拟器并非易事[54, 69]。因此，通过为推荐场景设计定制化模块来模拟人类行为，从而释放由 LLM 赋能的生成式智能体的潜力，已成为关键研究重点。

为弥合这一差距，我们提出 Agent4Rec——一个面向推荐场景的通用用户模拟器，它由两个核心方面构成：LLM 赋能的生成式智能体与推荐环境（参见图 2）。从用户视角出发，我们在每个推荐场景中模拟 1,000 个由 LLM 赋能的生成式智能体；每个智能体均基于真实世界数据集进行初始化，并由三个关键模块组成：用户画像、记忆与行动模块。画像模块充当一个存储库，用于

个性化的社会特质与历史偏好[38]，促进用户画像与真实人类性格特征的对齐。记忆模块以自然语言记录过去的观看行为、系统交互以及情绪记忆（即用户感受与疲劳水平），从而以连贯的方式实现信息检索、偏好累积与情绪驱动的反思。行动模块使这些智能体能够直接与推荐环境交互，包括品味驱动的行动（即观看或忽略推荐电影、评分、生成观影后的感受），以及情绪驱动的行动（即退出系统、评估推荐列表、表达可被人类理解的评论）。从推荐系统仿真的角度看，物品由预先设定的推荐算法进行推荐，并以逐页的格式排序，以模拟真实世界的推荐平台。一方面，我们的模拟器主要集成基于协同过滤的推荐策略，包括随机、最热门、矩阵分解（MF）[24]、LightGCN[16]和MultVAE[28]。另一方面，我们以可扩展性为核心原则来构建该模拟器。通过提供开放接口，我们使研究人员与从业者能够轻松部署其选择的任意推荐算法。

为了系统地评估我们提出的 Agent4Rec 的有效性与局限性，我们从用户视角与推荐系统视角两方面开展了全面实验。从用户视角出发，我们主要关注智能体对齐程度的评估。具体而言，我们使用多种指标与统计检验来评估智能体在多大程度上能够确保真实用户的社会特征、个性与偏好的一致性。在推荐系统仿真方面，我们评估了采用不同算法配置的多种推荐器。我们的评估指标覆盖多个维度，包括用户被推荐并观看的平均物品数量、平均用户评分、用户参与时长以及整体用户满意度。与此同时，从模拟器收集的智能体反馈作为增强数据，用于迭代训练并改进推荐策略。随后，我们使用标准化的离线指标评估这种由反馈驱动的推荐器增强效果。这种将仿真反馈与传统离线评估相结合的双重方法，确保了对推荐算法的全面评估。

为探索该模拟器在研究推荐任务中尚未解决挑战方面的潜力，我们开展了两项实验。在第一项实验中，我们在模拟器内模拟过滤气泡效应——一种用户持续接触相似或相互强化内容的情境，从而导致物品属性多样性降低[10, 27]。该研究旨在理解反馈回路在多大程度上会放大此类中心化推荐现象。此外，我们将模拟器用作数据收集工具，以开创一种面向数据的因果发现方法[17, 62]。该方法生成了一个稳健的因果图，使我们能够揭示推荐系统中推断数据生成过程的复杂潜在因果关系[11]。

我们的主要贡献总结如下：

- We develop Agent4Rec, a general recommendation simulator utilizing LLM-empowered agents to emulate and deduce user-personalized preferences and behavior patterns. These agents, with their carefully designed modules tailored to recommendation, enable the emulation of human cognitive mechanisms.
- We delve into both the capabilities and limitations of Agent4Rec by conducting extensive evaluations for generative agent-based simulation in recommender systems. We employ statistical metrics and tests for user alignment evaluation and propose a dual parallel evaluation considering both offline performance and simulation feedback.
- Using Agent4Rec as a data collection tool, we replicate a pervasive issue — the filter bubble effect—and unveil the underlying causal relationships embedded within recommender system scenarios. These observations showcase the potential of Agent4Rec to inspire new research directions.

We believe that Agent4Rec stands at the intersection of cutting-edge technology and the challenges of recommender systems, offering an experimental platform for insights that will inspire more work in this research direction.

2 AGENT4REC

Agent4Rec, as a user simulation in recommendation scenarios, is expected to accurately mirror user behaviors, effectively forecast long-term user preferences, and systematically evaluate recommendation algorithms by leveraging the human-like capabilities of LLM-empowered generative agents. To achieve this goal, two core facets are considered: (1) designing agent architectures that faithfully mimic user personalized preferences and human cognitive reasoning, and (2) constructing a recommendation environment that ensures its reliability, extensibility, and adaptability. Figure 2 demonstrated the overview of Agent4Rec's framework, which is developed by modifying LangChain, with all agents being powered by the gpt-3.5-turbo version of ChatGPT.

Task Formulation. Given a user $u \in \mathcal{U}$ and an item $i \in \mathcal{I}$, let $y_{ui} = 1$ denote that user u has interacted with item i, and subsequently rated it with $r_{ui} \in \{1, 2, 3, 4, 5\}$. Conversely, $y_{ui} = 0$ indicates that the user has not adopted the item. The quality of each item i can be represented by $R_i \doteq \frac{1}{\sum_{u\in\mathcal{U}} y_{ui}} \sum_{u\in\mathcal{U}} y_{ui} \cdot r_{ui}$, while its popularity is denoted by P_i. Additionally, the genre set of the item is given by $G_i \subset \mathcal{G}$. The simulator's overarching goal is to faithfully distill the human genuine preferences such as $\hat{y}_{ui}$ and $\hat{r}_{ui}$ of user u for an unseen recommended item i.

2.1 Agent Architecture

Generative agents in Agent4Rec, utilizing LLM as its foundational architecture, refine their capabilities with three specialized modules tailored for recommendation scenarios: a profile module, a memory module, and an action module. Specifically, to emulate personalized genuine human behaviors, each agent integrates a user profile module to reflect individualized social traits and preferences. Furthermore, drawing inspiration from human cognitive processes, agents are equipped with memory and action modules, enabling them to store, retrieve, and apply past interactions and emotions to generate behaviors in a coherent manner.

2.1.1 *Profile Module.*

In the domain of personalized recommendation simulation, the user profile module stands as a cornerstone, playing a crucial role in the efficacy of agents' alignment with genuine human behaviors. To lay a reliable foundation for the generative agent's subsequent simulations and evaluations, the benchmark dataset (*e.g.,* MovieLens-1M [15], Steam [22], Amazon-Book [37]) is used for initialization. Each agent's profile contains two components: social traits and unique tastes [33, 68].

Social traits encompass three key traits capturing the element of an individual's personality and characteristics in recommendation scenarios, that is activity, conformity, and diversity. Activity quantifies the frequency and breadth of a user's interactions with recommended items, delineating between users who extensively watch and rate a number of items and those who confine themselves to a minimal set [38]. The activity trait for user u can be mathematically articulated as: $T^u_{act} \doteq \sum_{i\in\mathcal{I}} y_{ui}$. Conformity delves into how closely a user's ratings align with average item ratings, drawing a distinction between users with unique perspectives and those whose opinions closely mirror popular sentiments [3, 70]. For user u, the conformity trait is defined as: $T^u_{conf} \doteq \frac{1}{\sum_{i\in\mathcal{I}} y_{ui}} \sum_{i\in\mathcal{I}} y_{ui} \cdot |r_{ui} - R_i|^2$. Diversity reflects the user's proclivity toward a diverse range of item genres or their inclination toward specific genres [2]. The diversity trait for user u is formulated as: $T^u_{div} \doteq |\cup_{i\in\{y_{ui}=1\}} G_i|$. Users typically exhibit specific distributions among these social traits (e.g., the long-tail distribution of user activity [66]). Accordingly, we segment them into three uneven tiers based on each respective trait.

To encode users' personalized preferences in natural language, we randomly select 25 items for each user from their viewing history. Items rated 3 or above are categorized as 'like' by the user, while those rated below 3 are deemed 'dislike'. Leveraging ChatGPT, we then distill and summarize the unique tastes and rating patterns the user exhibited. These personalized item tastes are incorporated as the second component into user profiles.

We underscore that in Agent4Rec, certain personal identifiers, such as name, gender, age, and occupation, are intentionally obscured to guarantee widespread applicability and address privacy concerns [7]. Although these attributes may be instrumental in shaping other types of agents, within the realm of recommendation, they do not dominate users' item preferences. Such preferences can be adequately deduced from historical viewing records, rating patterns, and insights embedded in user interactions [25].

2.1.2 *Memory Module.*

Humans retain diverse memories, bifurcating mainly into factual and emotional categories. Of these, emotional memories constitute the core of personal history and exert a stronger influence on decision-making [26]. Although pioneering studies on agents have detailed the architecture of memory, providing foundational blueprints for subsequent explorations, the emotional memories have been largely overlooked [42, 54].

In our Agent4Rec, we embed a specialized memory module within each generative agent, logging both factual and emotional memories. Tailored for the recommendation task, factual memories encapsulate interactive behaviors within the recommender system, while emotional memories capture the psychological feeling stemming from these interactions. Specifically, factual memory mainly contains the list of recommended items, along with user feedback.

• 我们开发了 Agent4Rec，这是一种通用的推荐模拟器，利用由 LLM 赋能的智能体来模拟并推断用户个性化偏好与行为模式。这些智能体配备了面向推荐任务精心设计的模块，从而能够模拟人类的认知机制。• 我们通过在推荐系统中对基于生成式智能体的模拟开展广泛评估，深入探讨 Agent4Rec 的能力与局限。我们采用统计指标与检验进行用户对齐评估，并提出一种双并行评估方案，同时考虑离线性能与模拟反馈。• 将 Agent4Rec 作为数据收集工具，我们复现了一个普遍存在的问题——过滤气泡效应，并揭示了嵌入在推荐系统场景中的潜在因果关系。这些观察结果展示了 Agent4Rec 激发新研究方向的潜力。

我们相信，Agent4Rec 处于前沿技术与推荐系统挑战的交汇点，提供了一个实验平台，用于产出洞见，从而激发在这一研究方向上的更多工作。

2 AGENT4REC

Agent4Rec 作为推荐场景中的用户模拟器，旨在准确映射用户行为，有效预测用户的长期偏好，并借助由 LLM 赋能的生成式智能体所具备的类人能力，对推荐算法进行系统性评估。为实现这一目标，我们考虑了两个核心方面：（1）设计能够忠实模拟用户个性化偏好与人类认知推理的智能体架构；（2）构建一个能够确保其可靠性、可扩展性与适应性的推荐环境。图 2 展示了 Agent4Rec 框架的整体概览，该框架通过对 LangChain 进行修改而开发，且所有智能体均由 ChatGPT 的 gpt-3.5-turbo 版本驱动。

任务表述。给定用户 æ ∈ $\mathcal{U}$ 和物品 ɑ 怀 $\mathcal{I}$，令 æ 懔贬蘐1 表示用户 æ 与物品 ɑ 恢发生过交互，并随后以 ɑ 贬蘐 {1, 2, 3, 4, 5} 对其进行评分。相反，æ 懔贬蘐0 表示用户尚未采纳该物品。每个物品 ɑ 怖的质量可由 ɐ 滄嘀 1 $\sum_{u\in\mathcal{U}}$ æ 怀 Σ æ $_{\in\mathcal{U}}$ æ 怀 æ ⇔嘀ɑ 贬蘐表示，而其流行度记为 ɐ 丶嘰 ɫ 揵 楔쉽葽笁型集合记为 a 嘀 $\mathcal{G}$。模拟器的总体目标是忠实地提炼用户 æ 对未见推荐物品 ɑ 怖的人类真实偏好，例如 ˜æ 懔贬蘐和 ˆɑ 戀 ɑ 抳

2.1 智能体架构

Agent4Rec 中的生成式智能体以 LLM 作为其基础架构，并通过三个面向推荐场景的专用模块来增强其能力：画像模块、记忆模块和行动模块。具体而言，为了模拟个性化且真实的人类行为，每个智能体都集成了用户画像模块，用以体现个体化的社会特征与偏好。此外，受人类认知过程的启发，智能体还配备了记忆与行动模块，使其能够存储、检索并运用过往的交互与情绪，从而以连贯的方式生成行为。

2.1.1 画像模块。在个性化推荐仿真领域，用户画像模块是基石，在代理与真实人类行为对齐的有效性方面发挥着关键作用。为生成式代理后续的仿真与评估奠定可靠基础，使用基准数据集（例如 MovieLens-1M [15]、Steam [22]、Amazon-Book [37]）进行初始化。每个代理的画像包含两个组成部分：社会特征与独特偏好 [33, 68]。

社会特征涵盖推荐场景中刻画个体人格与特性的三个关键特征，即活跃度、一致性和多样性。活跃度量化用户与被推荐物品交互的频率与广度，用以区分那些大量观看并评价多种物品的用户与仅局限于极少数物品集合的用户[38]。用户 æ 訔귁跃度特征可用数学形式表示为：ɐ 燔贬茹꾐 ː æ Σ ɑ 怀 æ 懔贬蘐。一致性探讨用户评分与物品平均评分的贴合程度，从而区分具有独特视角的用户与其观点高度贴近大众情绪的用户[3, 70]。对于用户 æ 㕾 蠘昭闸 铣귁 讛裝꾐 ɑ 쇰熄蓇 $\sum_{i\in\mathcal{I}}$ æ 怀 $_{ui}$ Σ ɑ 怀 æ 懔贬蘐 ɑ 贬蘐 ɐ 滄嘀。多样性反映用户对多种物品类型的偏好倾向，或其对特定类型的偏好倾向[2]。用户 æ 的多样性特征表述为：ɐ 燔贬蓟꾗 ː æ |∪ ɑ 怀 æ 怀 $_{ui}$ =1} a 嘀 用户通常在这些社会特征上呈现特定的分布（例如用户活跃度的长尾分布[66]）。因此，我们依据每个特征分别将用户划分为三个不均衡的层级。

为了用自然语言编码用户的个性化偏好，我们从每位用户的观看历史中随机选择 25 个条目。评分为 3 或以上的条目被归类为用户“喜欢”，而评分低于 3 的条目则被视为“不喜欢”。借助 ChatGPT，我们进一步提炼并总结用户所表现出的独特口味与评分模式。这些个性化的条目偏好作为第二个组成部分被纳入用户画像中。

我们强调，在 Agent4Rec 中，某些个人身份标识（如姓名、性别、年龄和职业）被有意遮蔽，以确保广泛适用性并解决隐私顾虑 [7]。尽管这些属性可能在塑造其他类型的智能体时发挥重要作用，但在推荐领域，它们并不主导用户对物品的偏好。此类偏好可以从历史观看记录、评分模式以及蕴含于用户交互中的洞见中得到充分推断 [25]。

2.1.2 记忆模块。人类保留着多样的记忆，主要分化为事实记忆和情感记忆两类。其中，情感记忆构成个人历史的核心，并对决策产生更强的影响[26]。尽管关于智能体的开创性研究已详细阐述了记忆的架构，为后续探索提供了基础蓝图，但情感记忆在很大程度上一直被忽视[42, 54]。

在我们的 Agent4Rec 中，我们在每个生成式智能体内嵌入了一个专门的记忆模块，用于记录事实记忆与情感记忆。针对推荐任务，事实记忆概括了推荐系统中的交互行为，而情感记忆则捕捉由这些交互引发的心理感受。具体而言，事实记忆主要包含推荐物品列表以及用户反馈。

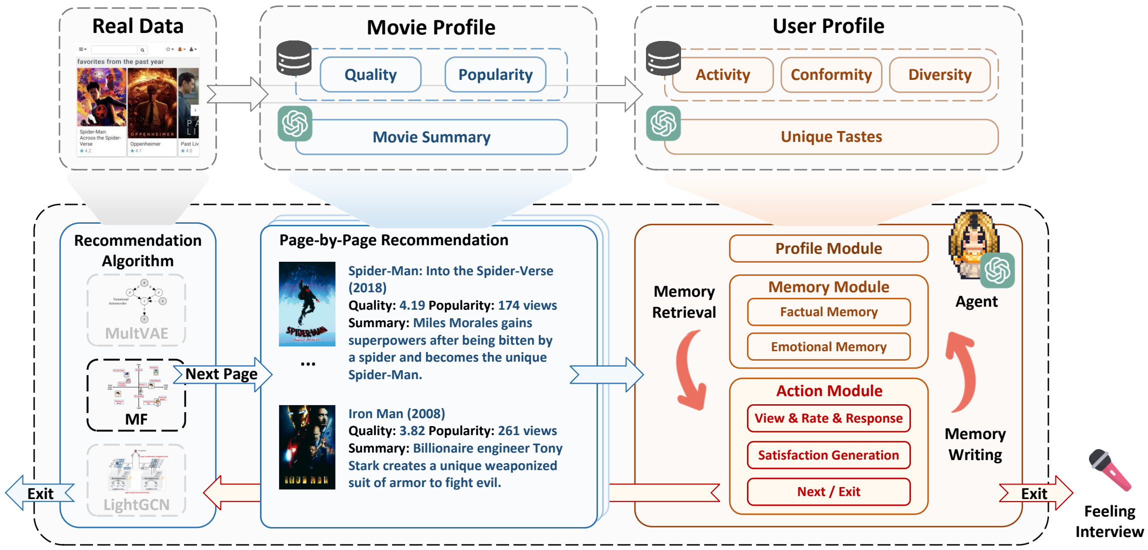


Figure 2: The overall framework of Agent4Rec. Our simulator consists of two core facets: LLM-empowered Generative Agents (Red Section) and Recommendation Environment (Blue Section). Both user and item profiles are initialized using real-world datasets (*e.g.,* MovieLens-1M, Steam, Amazon-Book). The recommendation algorithm, an adaptable component of the system, generates item recommendations presented to the agents in a page-by-page manner. Agents, enhanced with specialized memory and action modules tailored for recommendation scenarios, simulate a wide range of behaviors including viewing, rating items, providing feedback, navigating to the next page, exiting the system, and participating in post-exit interviews.

The feedback covers aspects like whether the user watches the item, his corresponding ratings, and potential exit behaviors. The emotional memory, on the other hand, records user feelings during system interactions, such as levels of fatigue and overall satisfaction. We aim to ensure the generative agent does not merely react based on past factual interactions but also takes into account the feelings, thereby mirroring genuine human behaviors more closely.

We store memories in two formats: natural language descriptions and vector representations. The former is designed for easy understanding by humans [45]. While vector representations are primed for efficient memory retrieval and extraction [57].

To help agents interact with the recommendation environment, we introduce three memory operations: memory retrieval, memory writing, and memory reflection.

- Memory Retrieval: Grounded in insights from studies [31, 59], this operation assists the agent in distilling the most relevant information from its memory module.
- Memory Writing: This operation enables the recording of agent-simulated interactions and emotions into the memory stream.
- Memory Reflection: Recognizing the influence of emotions over user behaviors in recommendations, we incorporate an emotion-driven self-reflection mechanism. This stands in contrast to conventional agent memory designs, such as self-summarization [36], self-verification [48], and self-correction [49], which predominantly condense or deduce advanced factual knowledge, often sidelining emotional feelings. Here, once the agent's actions surpass a pre-defined count, it triggers a reflection process. Armed with the prowess of the LLM, the agent introspects its satisfaction with the recommendations and assesses its fatigue levels, offering a deeper understanding of its cognitive state.

2.1.3 ***Action Module.*** Equipping agents with user profiles and memory modules enables them to exhibit diverse actions akin to humans based on current observations [58]. In Agent4Rec, we design an action module specifically tailored for recommendation domain, which encompasses two broad categories of actions:

- Taste-driven Actions: view, rate, and generate post-viewing feelings for items. In Agent4Rec, the recommended items are first generated by recommendation algorithms and then presented to agents in a page-by-page manner (further details are available in Section 2.2). Guided by their taste, agents assess each item on the page for consistency with their preferences. They may choose to watch certain items that pique their interest while bypassing others and subsequently providing ratings and feelings for each item they watch.
- Emotion-driven Actions: exit and rate recommender systems, and do post-exit interviews. Emotions in the recommendation environment can shape an agent's experience significantly, which are often overlooked in simulations [46, 54]. An agent's satisfaction with previously recommended items and its current fatigue level influences its decision to continue exploring further recommendation pages or to exit the recommender system. To better simulate this multifaceted decision-making, we enhance the agent's ability for emotional reasoning via Chain-of-Thought [56]. Initially, the agent discerns the current recommendation page and retrieves its satisfaction level with preceding recommendations from its emotional memory. Following this, the agent autonomously expresses its satisfaction and fatigue level for the current recommendation page. Drawing upon these insights, combined with its personalized activity trait, the agent decides whether to exit the system. Post-exit, we conduct an interview with each agent, aiming at

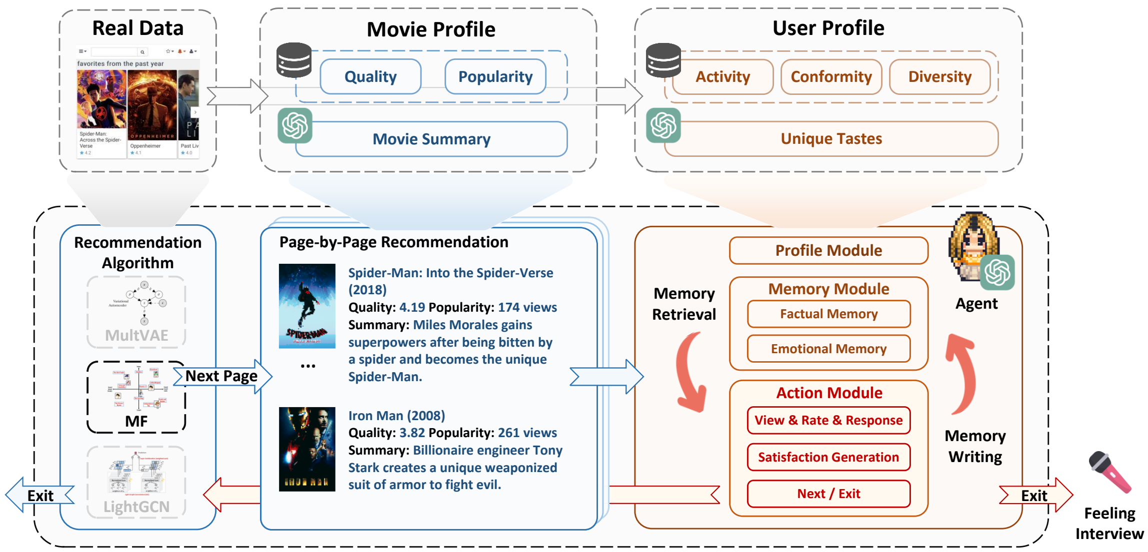


图2：Agent4Rec 的整体框架。我们的模拟器由两个核心方面构成：LLM 赋能的生成式智能体（红色部分）与推荐环境（蓝色部分）。用户与物品画像均使用真实世界数据集（例如 MovieLens-1M、Steam、Amazon-Book）进行初始化。推荐算法作为系统的可适配组件，生成物品推荐，并以逐页的方式呈现给智能体。智能体通过面向推荐场景定制的专用记忆与行动模块得到增强，可模拟包括浏览、评分、提供反馈、跳转到下一页、退出系统以及参与退出后访谈在内的多种行为。

反馈涵盖了诸如用户是否观看该条目、其对应评分以及潜在退出行为等方面。另一方面，情感记忆记录了用户在与系统交互过程中的感受，例如疲劳程度和整体满意度。我们旨在确保生成式智能体不仅仅基于过去的事实性交互作出反应，还会将这些感受纳入考量，从而更贴近地映射真实的人类行为。

我们以两种格式存储记忆：自然语言描述和向量表示。前者旨在便于人类理解[45]，而向量表示则为高效的记忆检索与提取而优化[57]。

为了帮助智能体与推荐环境交互，我们引入三种记忆操作：记忆检索、记忆写入和记忆反思。

• 记忆检索：基于研究[31, 59]的洞见，该操作帮助智能体从其记忆模块中提炼出最相关的信息。• 记忆写入：该操作使得将智能体模拟的交互与情绪记录到记忆流中成为可能。• 记忆反思：鉴于情绪对推荐中用户行为的影响，我们引入一种情绪驱动的自我反思机制。这与传统的智能体记忆设计形成对比，例如自我总结[36]、自我验证[48]和自我纠正[49]，这些方法主要压缩或推导高级事实性知识，往往忽视情绪感受。在此，当智能体的行动次数超过预先设定的阈值时，将触发反思过程。借助LLM的强大能力，智能体会内省其对推荐的满意度并评估其疲劳水平，从而更深入地理解其认知状态。

2.1.3 行动模块。为智能体配备用户画像和记忆模块，使其能够基于当前观测展现出类似人类的多样化行动 [58]。在 Agent4Rec 中，我们设计了一个专门面向推荐领域的行动模块，涵盖两大类行动：

• 品味驱动的动作：对条目进行查看、评分，并生成观看后的感受。在 Agent4Rec 中，推荐条目首先由推荐算法生成，然后以逐页的方式呈现给智能体（更多细节见第 2.2 节）。在品味的引导下，智能体会评估页面上的每个条目与其偏好的一致性。他们可能会选择观看某些引起兴趣的条目，同时跳过其他条目，并随后为其观看的每个条目提供评分和感受。

• 情绪驱动的行为：退出并对推荐系统进行评分，以及在退出后进行访谈。推荐环境中的情绪会显著塑造智能体的体验，但在仿真中常常被忽视[46, 54]。智能体对先前推荐物品的满意度以及其当前的疲劳水平，会影响其决定是继续浏览更多推荐页面，还是退出推荐系统。为更好地模拟这种多维度的决策过程，我们通过思维链（Chain-of-Thought）[56]增强智能体的情绪推理能力。起初，智能体识别当前的推荐页面，并从其情绪记忆中检索其对先前推荐的满意度水平。随后，智能体自主表达其对当前推荐页面的满意度与疲劳水平。基于这些洞见，并结合其个性化的活动特质，智能体决定是否退出系统。退出后，我们对每个智能体进行访谈，旨在

capturing agents' ratings and overall impressions of the recommender system, offering explicit explanations of their behaviors while navigating the system. This interview-style feedback provides a richer and more human-understandable evaluation of the system, enhancing the insights from traditional metrics. An in-depth exploration of the interview can be found in Section 3.5.

2.2 Recommendation Environment

Agent4Rec simulates the interactions between agents and the recommendation environment. We discuss three aspects of environment construction that resonate with real-world scenarios, including item profile generation, page-by-page recommendation scenarios, and recommendation algorithm designs.

- Item Profile Generation: We construct item profiles to capture key item features, including quality, popularity, genre, and summary. Quality is deduced from historical ratings, popularity is based on the number of reviews, while genre and summary are generated by LLM. Our goal goes beyond encapsulating the uniqueness of the item in a profile to simulate the recommendation scene of real users. We also aim to test whether LLM has potential hallucinations regarding the item. Our approach utilizes a few-shot learning approach, tasking the LLM with classifying the item into one of 18 genres and generating a summary using only the item title. If the LLM's genre classification aligns with the data, it signifies its knowledge of the item. To maintain reliability, items causing genre misclassification by the LLM are pruned, reducing hallucination risks. This approach ensures the agent's trustworthiness in simulating user behavior.
- Page-by-Page Recommendation Scenario: Our simulator mirrors the operation of real-world recommendation platforms like Netflix, YouTube, and Douban, functioning in a page-by-page manner. Users are initially presented with a list of item recommendations on each page. Based on interactions, preferences, and feedback, subsequent pages could be set to tailor the recommendations further, aiming for a more refined user experience. For further details and experimental results, please refer to Section 3.4.
- Recommendation Algorithm Designs: In Agent4Rec, the recommendation algorithm is structured as a standalone module, with a core focus on extensibility. This design encompasses pre-implemented collaborative filtering-based strategies, including random, most popular, Matrix Factorization (MF) [24], LightGCN [16], and MultVAE [28]. Moreover, it incorporates an open interface, enabling researchers and practitioners to effortlessly integrate external recommendation algorithms. This adaptability ensures that Agent4Rec could be a versatile platform for comprehensive evaluations and the collection of valuable user feedback in the future.

3 AGENT ALIGNMENT EVALUATION.

With the specialized simulator, Agent4Rec, tailored for recommendations, we would like to explore an essential research question:

- **RQ1:** To what extent can LLM-empowered generative agents truly simulate the behaviors of genuine, independent humans in recommender systems?

In this section, we will delve into the capability and limitations of generative agents from various perspectives, including the alignment of user behavior (such as user taste, rating distribution, and social traits) and the evaluation of the recommendation environment (including recommendation strategies evaluation, page-by-page recommendation enhancements, and the case study of interview).

Motivation. To appropriately respond to recommended items, generative agents need to have a clear understanding of their own preferences. We conjecture that an independent, personalized agent, initialized from real users in MovieLens-1M, should maintain long-term preference coherence. In practice, this implies that the agent should be adept at distinguishing the items that real users favor.

Setting. To validate how well generative agents align with the preferences encoded in their user profiles, we task agents with distinguishing between the items that the corresponding real users have interacted with and those they have not. We conduct experiments on three real-world datasets (*i.e.,* Movielens-1M [15], Steam[22], Amazon-Book[37]). Specifically, a total of 1,000 agents will each be randomly assigned 20 items. Among these, the ratio between items the user has interacted with (*i.e.,* $y_{ui} = 1$) but was not utilized for profile initialization and items the user has not interacted with (*i.e.,* $y_{ui} = 0$) is set as $1 : m$, with $m \in \{1, 2, 3, 9\}$. Under this setting, agent responses (*i.e.,* $\hat{y}_{ui}$) to recommended items are considered binary discrimination, taking values between 0 and 1.

Results. Table 1 reports the empirical discrimination results across various metrics. The best performance for each metric is highlighted in bold and marked with an asterisk. We observe that:

- **Generative agents consistently and impressively identify items aligned with user preferences.** Specifically, regardless of the number of distracting items introduced, agents maintain a high accuracy of around 65% and recall of about 75%. This high fidelity can be attributed to the personalized profile faithfully mirroring the user's genuine interests. It further reflects that the agents effectively encapsulate a substantial portion of the real preferences, signifying the feasibility of LLM-powered generative agents in recommendation simulation.
- **Agents tend to maintain a relatively consistent count of preferred items, a phenomenon potentially stemming from inherent hallucinations in the LLM.** Notably, while Accuracy and Recall show satisfactory results, metrics like Precision and F1 Score, which emphasize the cost of false positives, experience a sharp decline from nearly 70% to about one quarter as the proportion of user-liked items decreases. We attribute this failure to LLM's inherent hallucinations that agents tend to consistently pick a set number of items. Such behavior highlights challenges in designing a more reliable recommendation simulator using LLM-empowered generative agents. However, we emphasize that in the subsequent simulation results with recommendation algorithms, a substantial proportion of recommended items align with user preferences, thereby endorsing high trustworthiness in those simulation outcomes.

3.1 Rating Distribution Alignment

Motivation. Beyond ensuring user-aligned behaviors at a micro-level for each agent, a comprehensive evaluation of the simulator requires that the generative agents accurately mirror real-world

捕捉智能体对推荐系统的评分与整体印象，并在其浏览系统的过程中对其行为给出明确解释。这种访谈式反馈提供了更丰富且更易于人类理解的系统评估，增强了传统指标所带来的洞见。关于该访谈的深入探讨见第 3.5 节。

2.2 推荐环境

Agent4Rec 模拟智能体与推荐环境之间的交互。我们讨论了与真实世界场景相契合的环境构建的三个方面，包括物品画像生成、逐页推荐场景以及推荐算法设计。

• 物品画像生成：我们构建物品画像以捕捉物品的关键特征，包括质量、热度、类型和摘要。质量由历史评分推断，热度基于评论数量，而类型与摘要由 LLM 生成。我们的目标不仅是通过画像概括物品的独特性，还要模拟真实用户的推荐场景。我们也旨在测试 LLM 是否会对该物品产生潜在幻觉。我们采用少样本学习方法，仅使用物品标题，要求 LLM 将物品归类到 18 种类型之一并生成摘要。如果 LLM 的类型分类与数据一致，则表明其对该物品具备相关知识。为保持可靠性，会剪除那些导致 LLM 类型误分类的物品，从而降低幻觉风险。该方法确保智能体在模拟用户行为时的可信度。

• 逐页推荐场景：我们的模拟器复现了现实世界推荐平台（如 Netflix、YouTube 和 豆瓣）的运行方式，以逐页形式工作。用户在每一页上最初会看到一份物品推荐列表。基于交互、偏好与反馈，后续页面可以进一步定制推荐，以实现更精细的用户体验。更多细节与实验结果请参见第 3.4 节。

• 推荐算法设计：在 Agent4Rec 中，推荐算法被构建为一个独立模块，核心关注点是可扩展性。该设计涵盖了预先实现的基于协同过滤的策略，包括随机、最热门、矩阵分解（MF）[24]、Light-GCN [16] 以及 MultVAE [28]。此外，它还提供了开放接口，使研究人员和从业者能够轻松集成外部推荐算法。这种适应性确保 Agent4Rec 在未来能够成为一个用于全面评估并收集有价值用户反馈的多功能平台。

3 智能体对齐评估。

借助专为推荐场景定制的专用模拟器 Agent4Rec，我们希望探讨一个关键的研究问题：

• RQ1：由 LLM 赋能的生成式智能体在推荐系统中究竟能在多大程度上真实地模拟真正、独立的人类行为？

在本节中，我们将从多个角度深入探讨生成式智能体的能力与局限性，包括用户行为的一致性（如用户偏好、评分分布以及社交特征）以及推荐环境的评估（包括推荐策略评估、逐页推荐增强，以及访谈案例研究）。

动机。为了对推荐物品做出恰当的响应，生成式智能体需要清晰地理解自身的偏好。我们推测，一个从 MovieLens-1M 的真实用户初始化的独立、个性化智能体，应当保持长期的偏好一致性。在实践中，这意味着该智能体应当善于区分真实用户所偏爱的物品。

设置。为了验证生成式智能体与其用户画像中编码的偏好对齐得有多好，我们让智能体执行一项任务：区分对应真实用户曾经交互过的物品与未交互过的物品。我们在三个真实世界数据集上进行实验（即 Movielens-1M [15]、Steam[22]、Amazon-Book[37]）。具体而言，共有 1,000 个智能体，每个将被随机分配 20 个物品。其中，用户已交互过（即 æ懔貥蘐1）但未用于画像初始化的物品，与用户未交互过的物品（即 æ懔貥蘐0）之间的比例设为 1：ᵅ 刊쇎擢큄 ∈ {1、2、3、9}。在该设置下，智能体对推荐物品的响应（即 æ懔貥蘐 被视为二元判别，取值在 0 和 1 之间。

结果。表 1 报告了在多种指标上的经验性区分结果。每个指标的最佳表现以粗体突出显示，并用星号标注。我们观察到：

• 生成式智能体能够持续且令人印象深刻地识别与用户偏好一致的条目。具体而言，无论引入多少干扰条目，智能体都能保持约 65% 的高准确率以及约 75% 的召回率。这种高保真度可归因于个性化画像忠实地映射了用户的真实兴趣。这也进一步表明，智能体有效地封装了相当一部分真实偏好，彰显了由 LLM 驱动的生成式智能体在推荐模拟中的可行性。

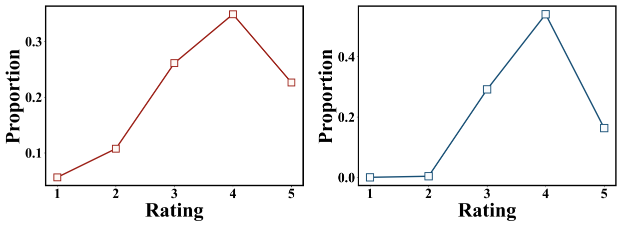
• 智能体往往会维持一个相对稳定的偏好物品数量，这一现象可能源于 LLM 的内在幻觉。值得注意的是，尽管 Accuracy 和 Recall 表现令人满意，但像 Precision 和 F1 Score 这类强调假阳性代价的指标，会随着用户喜欢物品占比的下降而急剧下滑，从接近 70% 降至约四分之一。我们将这一失败归因于 LLM 的内在幻觉：智能体倾向于始终选择固定数量的物品。这种行为凸显了在使用 LLM 赋能的生成式智能体来设计更可靠的推荐模拟器时所面临的挑战。然而，我们强调，在后续结合推荐算法的模拟结果中，相当比例的推荐物品与用户偏好一致，从而支持这些模拟结果具有较高的可信度。

3.1 评分分布对齐

动机。除了在每个智能体的微观层面确保与用户对齐的行为之外，对模拟器的全面评估还要求生成式智能体能够准确映射现实世界

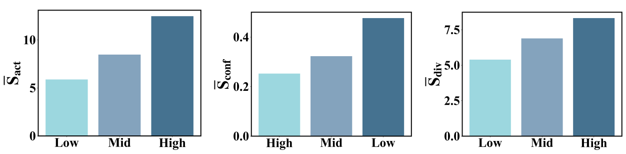
Table 1: User Taste Alignment across Movielens, Amazon-Book, and Steam datasets

	MovieLens				Amazon-Book				Steam			
1:m	Accuracy	Recall	Precision	F1 Score	Accuracy	Recall	Precision	F1 Score	Accuracy	Recall	Precision	F1 Score
1:1	**0.6912***	0.7460	**0.6914***	**0.6982***	**0.7190***	**0.7276***	**0.7335***	**0.7002***	**0.6892***	0.7059	**0.7031***	**0.6786***
1:2	0.6466	0.7602	0.5058	0.5874	0.6842	0.6888	0.5763	0.5850	0.6755	0.7316	0.5371	0.5950
1:3	0.6675	0.7623	0.4562	0.5433	0.6707	0.6909	0.4423	0.5098	0.6505	**0.7381***	0.4446	0.5194
1:9	0.6175	**0.7753***	0.2139	0.3232	0.6617	0.6939	0.2369	0.3183	0.6021	0.7213	0.1901	0.2822



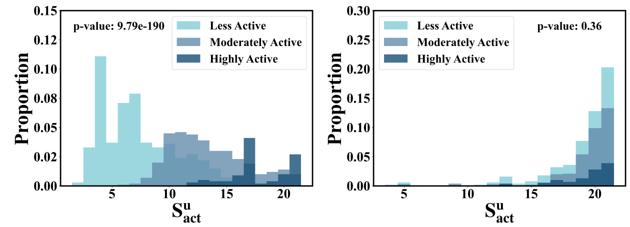
(a) Distribution on MovieLens (b) Agent-simulated distribution

Figure 3: Comparison between ground-truth and agent-simulated rating distributions.



(a) Activity (b) Conformity (c) Diversity

Figure 4: Averaged scores of activity, conformity, and diversity among agent groups with varying degrees of social traits.



(a) w Activity trait (b) w/o Activity trait

Figure 5: Distribution of interaction numbers among agents with varying levels of user activity.

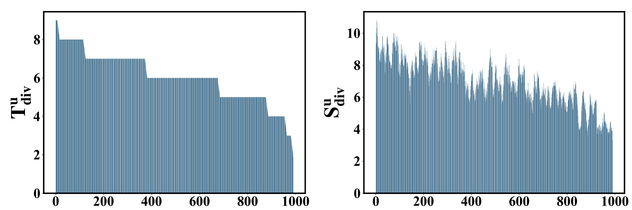

(a) Ground-truth diversity (b) Simulated diversity

Figure 6: Comparison between the individual level distributions of ground-truth and agent-simulated diversity scores.

user behavioral patterns at a macro scale. Specifically, in terms of rating distribution, the goal is to maintain consistency with MovieLens-1M data distributions.

Results. Figures (3a) and (3b) illustrate the ground-truth rating distribution in the Movielens-1M dataset and the distribution of ratings generated by agents, respectively. The figures demonstrate a strong alignment between the simulated rating distribution and the actual distribution. Specifically, ratings at 4 dominate the overall distribution, while low ratings (1-2) constitute only a small portion. It is worth mentioning that agents tend to give very few 1-2 ratings, which differs from genuine human behavior. This is attributed to the LLM's extensive prior knowledge of movies, as agents tend to avoid watching low-quality films in advance. Consequently, simulating the user action of giving low ratings after choosing to watch a movie becomes challenging. Moreover, Figure (3b) represents the simulation results under the MF algorithm, and similar trends are observed under other recommendation algorithms. This consistency across different algorithms validates the potential of Agent4Rec. We demonstrate through ablation that the social traits component of our agents contribute critically to the believability of agent behavior.

3.2 Social Traits Alignment

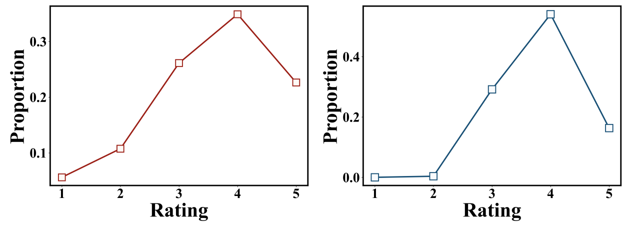
Motivation. In real-world recommendation scenarios, user behavior patterns are influenced by various factors, not solely determined by personal preferences but also impacted by social attributes such as activity, conformity, and diversity [38]. Accordingly, we specially designed a user profile for Agent4Rec that incorporates these social traits. Following the definitions in Section 2.1.1 and based on statistics from MovieLens, agents are categorized into three different levels (high, medium, and low) for each trait, giving a consistent prompt for agents within each tier. We conjecture that agents exhibiting similar preferences might still display differentiated behavior patterns based on their unique social traits.

Results. To validate the significance of the user profile module's design, it is essential to probe from multiple perspectives, including (1) whether agents categorized into different tiers exhibit distinct behaviors (Figure 4); (2) whether ablation studies can detect differences between agents with and without social traits in their profiles (Figure 5); and (3) whether the simulated traits distribution of agents aligns with that of actual users (Figure 6).

Extensive experimental results from distribution analysis, ablation studies, and statistical tests demonstrate the pivotal role of social traits in user profile construction, significantly influencing agent behavior. This further illustrates the universal applicability and referential value of Agent4Rec's setup, which is specifically designed for recommendation tasks. It's worth noting that the diversity trait might exhibit minimal distinction in agent behavior, possibly due to the strong overlap of movie categories in the MovieLens dataset, suggesting further research and validation with alternative datasets.

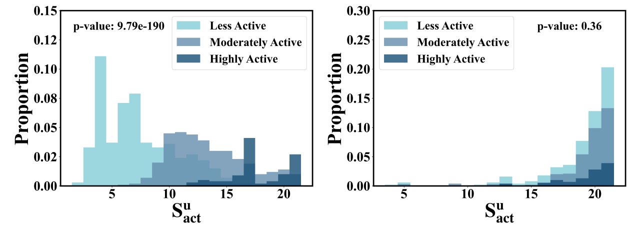
表 1：Movielens、Amazon-Book 和 Steam 数据集中的用户品味对齐情况

	源文本：**MovieLens**				**Amazon-Book**				**Steam**			译文：
1:m	Accuracy	Recall	Precision	F1 Score	Accuracy	Recall	Precision	F1 Score	Accuracy	Recall	Precision	F1 Score
1:1	**0.6912***	0.7460	**0.6914***	**0.6982***	**0.7190***	**0.7276***	**0.7335***	**0.7002***	**0.6892***	0.7059	**0.7031***	**0.6786***
1:2	0.6466	0.7602	0.5058	0.5874	0.6842	0.6888	0.5763	0.5850	0.6755	0.7316	0.5371	0.5950
1:3	0.6675	0.7623	0.4562	0.5433	0.6707	0.6909	0.4423	0.5098	0.6505	**0.7381***	0.4446	0.5194
1:9	0.6175	**0.7753***	0.2139	0.3232	0.6617	0.6939	0.2369	0.3183	0.6021	0.7213	0.1901	0.2822



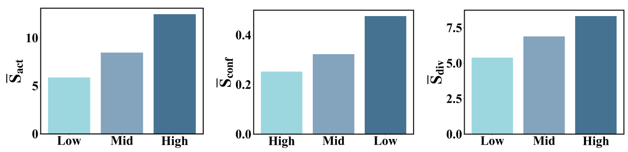
(a) MovieLens 上的分布（b）智能体模拟分布

图 3：真实值与智能体模拟评分分布的比较。



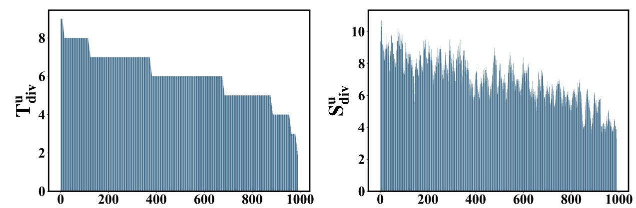
(a) w Activity trait **(b) w/o Activity trait**

图 5：不同用户活跃度水平的智能体之间交互次数的分布。



(a) Activity **(b) Conformity** **(c) Diversity**

图 4：在具有不同程度社会特质的智能体群体中，活动性、从众性和多样性的平均得分。



(a) Ground-truth diversity **(b) Simulated diversity**

图 6：真实值与智能体模拟的多样性得分在个体层面分布的比较。

宏观尺度上的用户行为模式。具体而言，在评分分布方面，目标是与 MovieLens-1M 数据分布保持一致。

结果。图（3a）和图（3b）分别展示了 Movielens-1M 数据集中真实评分的分布，以及由智能体生成的评分分布。图中表明，模拟评分分布与真实分布高度一致。具体而言，评分为 4 的占据整体分布的主导地位，而低评分（1-2）仅占很小一部分。值得一提的是，智能体倾向于给出极少的 1-2 评分，这与真实人类行为有所不同。这归因于 LLM 对电影的广泛先验知识，因为智能体往往会提前避免观看低质量影片。因此，在选择观看电影之后再给出低评分的用户行为更难以模拟。此外，图（3b）表示在 MF 算法下的仿真结果，在其他推荐算法下也观察到类似趋势。这种跨不同算法的一致性验证了 Agent4Rec 的潜力。我们通过消融实验表明，智能体的社会特质组件对智能体行为的可信度具有关键贡献。

3.2 社会特质对齐

动机。在真实世界的推荐场景中，用户行为模式受到多种因素的影响，并非仅由个人偏好决定，还会受到活跃度、从众性和多样性等社会属性的影响[38]。因此，我们专门为 Agent4Rec 设计了一个用户画像，将这些因素纳入其中。

社会特质。遵循第 2.1.1 节中的定义，并基于 MovieLens 的统计数据，我们将智能体在每项特质上划分为三个不同等级（高、中、低），从而为同一层级内的智能体提供一致的提示。我们推测，表现出相似偏好的智能体仍可能基于其独特的社会特质呈现出差异化的行为模式。

结果。为验证用户画像模块设计的重要性，有必要从多个角度进行探究，包括：（1）被划分到不同层级的智能体是否表现出不同的行为（图4）；（2）消融研究是否能够检测出画像中包含与不包含社会特质的智能体之间的差异（图5）；以及（3）智能体的模拟特质分布是否与真实用户的特质分布一致（图6）。

来自分布分析、消融研究和统计检验的大量实验结果表明，社会特质在用户画像构建中发挥着关键作用，并显著影响智能体行为。这进一步说明了 Agent4Rec 的设置具有普适适用性和参考价值，其专门为推荐任务而设计。值得注意的是，多样性特质在智能体行为中可能表现出较小差异，这可能是由于 MovieLens 数据集中电影类别高度重叠所致，提示需要在其他数据集上开展进一步研究与验证。

Table 2: Recommendation strategies evaluation.

	$\overline{P}_{view}$	$\overline{N}_{like}$	$\overline{P}_{like}$	$\overline{N}_{exit}$	$\overline{S}_{sat}$
Random	0.312	3.3	0.269	2.99	2.93
Pop	0.398	4.45	0.360	3.01	3.42
MF	0.488	**6.07***	0.462	**3.17***	3.80
MultVAE	0.495	5.69	0.452	3.10	3.75
LightGCN	**0.502***	5.73	**0.465***	3.02	**3.85***

3.3 Recommendation Strategy Evaluation

Motivation. Human users display varying levels of satisfaction with different recommendation Algorithms. For example, users generally gain higher satisfaction with advanced strategies, compared to random recommendations. Assuming generative agents can accurately simulate genuine human behaviors, it's plausible that these agents would exhibit similar satisfaction trends to those observations in humans.

Setting. For a fair comparison, we employ mainly collaborative filtering-based recommendation strategies (*i.e.,* random, most popular, MF [24], LightGCN [16], and MultVAE [28]) in our simulator and evaluate their satisfaction trends on MovieLens. In Agent4Rec, four movies are displayed on each recommendation page, and agents will take actions such as viewing and rating based on their personal preferences. Simultaneously, agents decide whether to proceed to the next recommended page or exit the recommendation system based on their satisfaction. The system enforces agents' exit after browsing a maximum of five pages. Once an agent exits, we request him to give a satisfaction score for the recommendation system, ranging from 1 to 10. Regarding rating higher than 3 as a signal of like, we collect the following multi-facets metrics after the whole simulation completes: average viewing ratio (*i.e.,* $\overline{P}_{view}$), the average number of likes (*i.e.,* $\overline{N}_{like}$), the average ratio of likes (*i.e.,* $\overline{P}_{like}$), the average number of exit page (*i.e.,* $\overline{N}_{exit}$), and the average user satisfaction score (*i.e.,* $\overline{S}_{sat}$).

Results. Table 2 reports the multi-facet satisfaction metrics for various recommendation strategies. It is clear that agents exhibit higher satisfaction with algorithm-based recommendations compared to random and popularity-based recommendations. This phenomenon aligns with observations in the real world, where well-designed recommendation algorithms can effectively address the issue of information overload in modern society, thereby enhancing users' online experience [5]. Furthermore, in line with prevailing insights in the research community, LightGCN outperforms both MF and MultVAE across diverse evaluation criteria. Such findings underscore the fine-grained evaluation capabilities of LLM-empowered agents for recommendation strategies. This further shed light on the potential of an agent-driven recommendation simulator for A/B testing, offering a more cost-efficient alternative to conventional online A/B testing.

3.4 Page-by-Page Recommendation Enhancement

Motivation. In real-world settings, recommendation platforms frequently collect immediate user behaviors to iteratively refine the recommender, aiming to accurately capture users' latest preferences.

Table 3: Page-by-page recommendation enhancement results over various algorithms.

	MF		MultVAE		LightGCN	
Offline	Recall	NDCG	Recall	NDCG	Recall	NDCG
Origin	0.1506	0.3561	0.1609	0.3512	0.1757	0.3937
+ Unviewed	0.1523	0.3557	0.1598	0.3487	0.1729	0.3849
+ Viewed	**0.1570***	**0.3604***	**0.1613***	**0.3540***	**0.1765***	**0.3943***
Simulation	$\overline{N}_{exit}$	$\overline{S}_{sat}$	$\overline{N}_{exit}$	$\overline{S}_{sat}$	$\overline{N}_{exit}$	$\overline{S}_{sat}$
Origin	3.17	3.80	3.10	3.75	3.02	3.85
+ Unviewed	3.03	3.77	3.01	3.77	3.06	3.81
+ Viewed	**3.27***	**3.83***	**3.18***	**3.87***	**3.10***	**3.92***

Our aim in designing the page-by-page recommendation setting is to emulate this feedback-driven recommendation enhancement.

Setting. After a complete recommendation simulation, we collect both viewed and unviewed movies for each agent. These two types of movies are then added as positive signals to the training set of each user to re-train the recommendation algorithms. We evaluate the performance of these retrained recommenders using standard offline metrics, such as Recall@20 and NDCG@20, as well as satisfaction simulation evaluations, namely $\overline{N}_{exit}$ and $\overline{S}_{sat}$.

Results. As depicted in Table 3, by leveraging movies viewed by the agent as augmented data, all recommendation algorithms exhibit improvements in both offline evaluation metrics and simulated satisfaction evaluations. However, when the training dataset is augmented with unviewed movies, the overall user experience typically deteriorates. Successfully emulating this feedback-driven recommendation augmentation indicates that movie choices by the agent can serve as a consistent indicator of a user's unique preferences.

3.5 Case Study of Feeling Interview

Motivation. Compared to conventional recommendation simulators [20, 50], the unique strength of LLM-empowered agent simulation lies in its ability to provide human-comprehensible explanations [19]. Eliciting explanations from agents provides insights into the reliability of simulations, enabling us to further refine the recommender system.

Results. Figure 7 presents a case from post-exit interviewing on MovieLens. The agent rates the recommended movies based on their personal tastes, social traits, and emotional memory. Specifically, the agent recognized that the recommender system did suggest movies in line with their preferences. However, there were aspects leading to less satisfaction. For instance, even though the agent has diverse interests, the system tended to recommend popular movies.

4 INSIGHTS AND EXPLORATION

Given Agent4Rec's promising simulation capabilities, we pose another profound research question:

- **RQ2:** Can Agent4Rec provide some insights on unresolved problems in the recommendation domain?

In this section, we discuss two insights drawn from our simulation results: replicating the filter bubble phenomenon and exploring causal discovery in movie recommendation tasks. We acknowledge

表 2：推荐策略评估。

源文本：	P_{view}	$\overline{N}_{like}$	$\overline{P}_{like}$	$\overline{N}_{exit}$	$\overline{S}_{sat}$
Random	0.312	3.3	0.269	2.99	2.93
Pop	0.398	4.45	0.360	3.01	3.42
MF	0.488	**6.07***	0.462	**3.17***	3.80
MultVAE	0.495	5.69	0.452	3.10	3.75
LightGCN	**0.502***	5.73	**0.465***	3.02	**3.85***

译文：

3.3 推荐策略评估

动机。人类用户对不同的推荐算法表现出不同程度的满意度。例如，与随机推荐相比，用户通常会对更先进的策略获得更高的满意度。假设生成式智能体能够准确模拟真实的人类行为，那么这些智能体很可能会表现出与人类观察结果相似的满意度趋势。

设置。为进行公平比较，我们在模拟器中主要采用基于协同过滤的推荐策略（即 random、most popular、MF [24]、LightGCN [16] 和 MultVAE [28]），并在 MovieLens 上评估它们的满意度趋势。在 Agent4Rec 中，每个推荐页面展示四部电影，智能体将基于其个人偏好采取诸如浏览与评分等行为。同时，智能体会根据其满意度决定是否继续进入下一页推荐或退出推荐系统。系统强制智能体在最多浏览五页后退出。一旦智能体退出，我们会要求其对推荐系统给出一个 1 到 10 的满意度评分。将评分高于 3 视为喜欢的信号，在整个仿真完成后，我们收集如下多维度指标：平均浏览率（即 ᐅ 丶挝蔄翾 æ 俐鑰 吭留뾜欢数量（即 ᐅ 牲蔄꾉 ɑ 鑰 吭留읐⚽薪뾜蔷 丶牲蔄꾉 ɑ 退、平均退出页数（即 ᐅ ɑ ⇔攝蔄虿），以及平均用户满意度得分（即 ᐅ 懔思茹虿）。

结果。表 2 报告了针对不同推荐策略的多维度满意度指标。可以清楚地看到，与随机推荐和基于流行度的推荐相比，智能体对基于算法的推荐表现出更高的满意度。这一现象与现实世界中的观察一致：设计良好的推荐算法能够有效缓解现代社会中的信息过载问题，从而提升用户的在线体验 [5]。此外，与研究界的主流认识一致，LightGCN 在多样化的评估标准上均优于 MF 和 MultVAE。这些发现凸显了由 LLM 赋能的智能体对推荐策略进行细粒度评估的能力。这也进一步揭示了由智能体驱动的推荐模拟器在 A/B 测试中的潜力，为传统的在线 A/B 测试提供了一种更具成本效益的替代方案。

3.4 逐页推荐增强

动机。在现实世界的场景中，推荐平台经常收集用户的即时行为，以迭代地改进推荐器，旨在准确捕捉用户的最新偏好。

表 3：各类算法在逐页推荐增强方面的结果。

Offline	MF Recall	MF NDCG	MultVAE Recall	MultVAE NDCG	LightGCN Recall	LightGCN NDCG
Origin	0.1506	0.3561	0.1609	0.3512	0.1757	0.3937
+ Unviewed	0.1523	0.3557	0.1598	0.3487	0.1729	0.3849
+ Viewed	**0.1570***	**0.3604***	**0.1613***	**0.3540***	**0.1765***	**0.3943***

Simulation	MF $\overline{N}_{exit}$	MF $\overline{S}_{sat}$	MultVAE $\overline{N}_{exit}$	MultVAE $\overline{S}_{sat}$	LightGCN $\overline{N}_{exit}$	LightGCN $\overline{S}_{sat}$
Origin	3.17	3.80	3.10	3.75	3.02	3.85
+ Unviewed	3.03	3.77	3.01	3.77	3.06	3.81
+ Viewed	**3.27***	**3.83***	**3.18***	**3.87***	**3.10***	**3.92***

我们在设计逐页推荐设置时的目标是模拟这种由反馈驱动的推荐增强。

设置。在完成一次完整的推荐模拟后，我们为每个智能体收集其已观看和未观看的电影。随后，将这两类电影作为正向信号加入到每个用户的训练集中，以重新训练推荐算法。我们使用标准的离线指标（如 Recall@20 和 NDCG@20）以及满意度模拟评估（即 ᐅ ɑ ⇔攝蔄虿和 ᐅ 懔思茹虿）来评估这些重新训练后的推荐器的性能。

结果。如表 3 所示，通过利用智能体观看过的电影作为增强数据，所有推荐算法在离线评估指标和模拟满意度评估中均表现出提升。然而，当使用未观看的电影对训练数据集进行增强时，整体用户体验通常会恶化。成功模拟这种由反馈驱动的推荐增强表明，智能体的电影选择可以作为用户独特偏好的一个一致性指标。

3.5 感受访谈案例研究

动机。与传统的推荐模拟器 [20, 50] 相比，LLM 赋能的智能体模拟的独特优势在于其能够提供人类可理解的解释 [19]。从智能体中引出解释可以为模拟的可靠性提供洞见，使我们能够进一步改进推荐系统。

结果。图 7 展示了一个来自 MovieLens 的退出后访谈案例。该智能体基于其个人品味、社会特质和情绪记忆对推荐电影进行评分。具体而言，智能体认识到推荐系统确实推荐了符合其偏好的电影。然而，也有一些方面导致其满意度较低。例如，尽管该智能体兴趣多样，系统仍倾向于推荐热门电影。

4 洞察与探索

鉴于 Agent4Rec 颇具前景的仿真能力，我们提出了另一个深刻的研究问题：

- RQ2：Agent4Rec 能否就推荐领域中尚未解决的问题提供一些见解？

在本节中，我们讨论从仿真结果中得到的两点见解：复现过滤气泡现象，以及在电影推荐任务中探索因果发现。我们承认

Case Study of Feeling Interview

Question: *Do you feel satisfied with the recommender system? Rate it from 1-10 and give an explanation.*
Rating: *6*
Reason: *While the recommender system did provide me with some movies that aligned with my taste, there were also a few recommendations that I disliked. The system took into account my personal preferences and historical ratings, which I appreciate as a Balanced Evaluator. However, as an Occasional Viewer, I am not easily swayed by movie recommendations and only seek out movies that strictly align with my taste. The system failed to consistently meet my preferences, resulting in a few unsatisfied memories. Additionally, as a Cinematic Trailblazer, I enjoy exploring unique and obscure movies, but the recommendations seemed to focus more on popular classics. Overall, the system had its strengths but fell short of fully satisfying my movie-watching preferences.*

Figure 7: One case study of post-exit feeling interview.

the difficulty of RQ2, and confine our discussion here to potential paths for future exploration.

4.1 Filter Bubble Effect

Motivation. The filter bubble effect is a pervasive challenge in recommender systems [12, 39]. This issue emerges when algorithm-based recommenders predict which movies users might prefer based on user feedback loops [35], resulting in increasingly homogeneous recommended contents. Our primary goal is to assess Agent4Rec's ability to replicate the filter bubble phenomenon.

Setting. To ensure fairness, we divide the MovieLens' movie pool into four equal parts, allowing the MF-based recommender to undergo four complete simulation rounds. In each round, recommended movies span up to 5 pages, with the MF-based recommender retrained after each simulation round. We evaluate the filter bubble effect based on content diversity at the individual user level. Two metrics are employed: $\overline{P}_{\text{top1-genre}}$, which represents the average percentage of top-1 genres among recommended movies, and $\overline{N}_{\text{genres}}$, indicating the average number of genres recommended in each simulation.

Results. Figure 8 reveals that as the number of iterations increases, movie recommendations tend to be more centralized. Specifically, the genre diversity, represented by $\overline{N}_{\text{genres}}$, decreases, while the dominance of the primary genre, denoted by $\overline{P}_{\text{top1-genre}}$, intensifies. This result further validates Agent4Rec's capability to reflect the filter bubble effect, an issue commonly observed in real-world recommender systems.

4.2 Discovering Causal Relationships

Motivation. Causal discovery aims to infer a causal structure, often represented as a causal graph, from observational data. This technique is crucial to comprehending the underlying mechanisms of specific fields. A recommendation simulator can aid researchers with data collection and in addressing latent confounding issues.

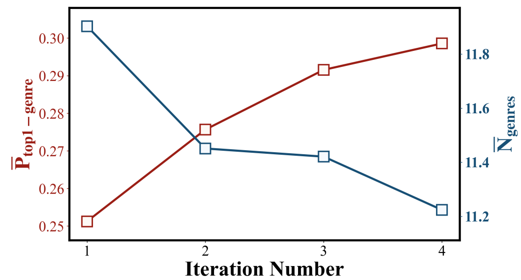


Figure 8: The simulation performance of Agent4Rec to emulate the filter bubble effect.

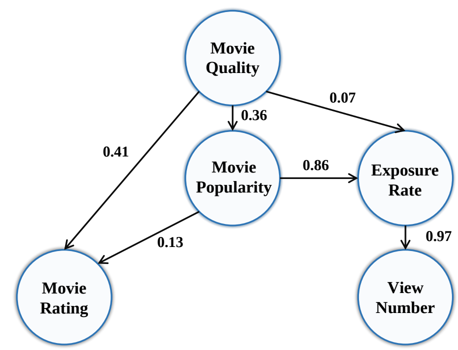


Figure 9: Learned causal graph among movie quality, popularity, exposure rate, view number, and movie rating.

In light of this, a question arises: can Agent4Rec be instrumental in uncovering causal relationships in recommender systems?

Setting. To understand the factors influencing movie ratings on MovieLens, for each movie, we collect data on four principal variables in addition to its rating simulated by agents: movie quality and popularity (sourced from the movie profile), exposure rate, and the number of times the movie is watched (sourced from the simulator). To probe the potential causal relationships within this simulated data, we employ the DirectLiNGAM algorithm [51]. This algorithm discovers a causal graph, *i.e.,* a weighted directed acyclic graph (DAG) in a linear system.

Results. Based on the learned causal graph in Figure 9, we can observe that:

- **Movie quality and movie popularity are the *causes* of movie ratings [43].** Although movie quality contributes most to agent ratings, agent ratings are also slightly influenced by the popularity of movies. This aligns with the inclination of humans to give high ratings to popular movies in real-world scenarios.
- **The feedback loop of amplifying the popularity bias is observed.** Highly popular movies receive increased exposure, resulting in more views by agents. When these popular items are introduced into the new training dataset, recommender systems are inclined to expose them more in the subsequent iterations, giving rise to what is known as popularity bias [11, 17, 65].

5 RELATED WORKS

In this section, we review two research lines of related works: LLM-empowered generative agents and recommender simulator.

感觉访谈案例研究

问题：你对推荐系统感到满意吗？请从 1-10 进行评分，并给出解释。

评分：6

原因：虽然推荐系统确实给我提供了一些符合我口味的电影，但也有几条推荐让我不喜欢。系统考虑了我的个人偏好和历史评分，作为一名平衡型评估者，我对此表示认可。然而，作为一名偶尔观影者，我不太容易被电影推荐所左右，只会去寻找那些严格符合我口味的电影。该系统未能持续满足我的偏好，导致留下了几段不太满意的观影记忆。此外，作为一名电影开拓者，我喜欢探索独特而冷门的电影，但这些推荐似乎更偏向于热门经典。总体而言，该系统有其优势，但未能完全满足我的观影偏好。

图 7：一项关于退出后感受访谈的案例研究。

RQ2 的难度，并将我们在此的讨论限定在未来探索的潜在路径上。

4.1 过滤气泡效应

动机。过滤气泡效应是推荐系统中普遍存在的挑战 [12, 39]。当基于算法的推荐器根据用户反馈回路预测用户可能偏好的电影时 [35]，就会出现这一问题，从而导致推荐内容日益同质化。我们的主要目标是评估 Agent4Rec 复现过滤气泡现象的能力。

设置。为确保公平性，我们将 MovieLens 的电影池划分为四个相等部分，使基于 MF 的推荐器能够经历四个完整的模拟轮次。在每一轮中，推荐电影最多覆盖 5 页，并在每个模拟轮次结束后对基于 MF 的推荐器进行重新训练。我们基于个体用户层面的内容多样性来评估过滤气泡效应。采用两个指标：ᵉ top1-genre，表示推荐电影中 top-1 类型的平均占比；以及 ᵉ genres，表示每次模拟中被推荐的类型平均数量。

—

结果。图 8 显示，随着迭代次数的增加，电影推荐趋于更加集中。具体而言，以 ᵉ genres 表示的类型多样性下降，而以 ᵉ top1-genre 表示的主导类型占比则增强。该结果进一步验证了 Agent4Rec 反映过滤气泡效应的能力，而过滤气泡效应是现实世界推荐系统中常见的问题。

4.2 发现因果关系

动机。因果发现旨在从观测数据中推断因果结构，通常以因果图的形式表示。这一技术对于理解特定领域的底层机制至关重要。推荐模拟器可以帮助研究人员进行数据收集，并解决潜在混杂问题。

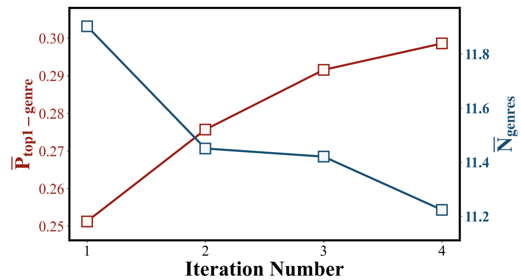


图 8：Agent4Rec 模拟过滤气泡效应的仿真性能。

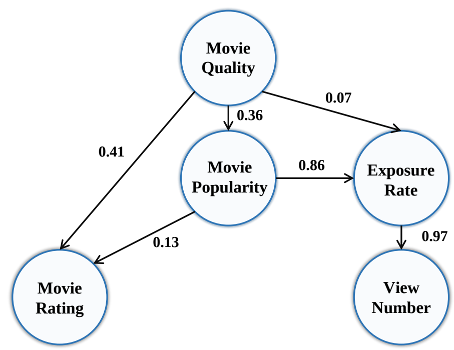


图 9：在电影质量、流行度、曝光率、观看次数和电影评分之间学习到的因果图。

鉴于此，一个问题随之而来：Agent4Rec 能否在揭示推荐系统中的因果关系方面发挥作用？

设定。为了理解影响 MovieLens 上电影评分的因素，我们针对每部电影，除由智能体模拟得到的评分外，还收集了四个主要变量的数据：电影质量与受欢迎程度（来自电影档案）、曝光率以及电影被观看的次数（来自模拟器）。为探究这些模拟数据中潜在的因果关系，我们采用 DirectLiNGAM 算法 [51]。该算法在一个线性系统中发现因果图，即一个加权有向无环图（DAG）。

结果。基于图 9 中学习到的因果图，我们可以观察到：

- 电影质量和电影受欢迎程度是电影评分的原因[43]。尽管电影质量对智能体评分的贡献最大，但智能体评分也会受到电影受欢迎程度的轻微影响。这与现实世界中人类倾向于给热门电影更高评分的现象一致。

- 观察到放大流行度偏差的反馈回路。高度热门的电影获得更多曝光，从而被智能体观看得更多。当这些热门项目被引入新的训练数据集时，推荐系统在后续迭代中倾向于进一步增加对它们的曝光，由此产生了所谓的流行度偏差 [11, 17, 65]。

5 相关工作

在本节中，我们回顾两条相关工作的研究脉络：LLM 赋能的生成式智能体与推荐系统模拟器。

5.1 LLM-empowered Generative Agents

AI agents are artificial entities that sense their environment, make decisions, and take actions [58]. Humans have long been pursuing intelligent agents to handle various tasks [8, 21, 41]. Recently, with the remarkable capability of Large Language Model (LLM) demonstrated, a substantial body of research has emerged [53]. Generative Agent [42] is a pioneer work that designs general agents equipped with memory, planning, and reflection abilities to simulate the human's daily life. Building upon this universal framework, the following agent architecture can be bifurcated into task-oriented agents and simulation-oriented agents [58].

The core objective of task-oriented agents is to execute predefined tasks established by humans [19, 34, 44, 52, 55, 57]. Within this category, Voyager [52] integrates LLM into the Minecraft universe, endowing the in-game character with the capacity to effectively explore the virtual world. ChatDev [44] introduces a novel agent collaboration paradigm in software development, harnessing group intelligence to streamline the development pipeline. AutoGen [57] further extends this paradigm by defining more complex roles and facilitating multi-agent communication to tackle a diverse range of challenges. In the realm of recommendation systems, RecMind [55] designs an LLM-powered autonomous recommender agent that possesses self-inspiring planning abilities and the capability to leverage external tools. InteRecAgent [19] takes a step beyond by enhancing LLM-based agents with integrated components, yielding recommenders that are both explainable and conversational.

Simulation-oriented agents are geared towards replicating human behaviors in specific scenarios, thereby enabling the acquisition of valuable data and the exploration of social issues [9, 30, 31, 42, 54, 60]. Among these agents, S^3 [9] leverages autonomous agents to simulate the dynamic evolution of public opinion on social platforms in response to trending social events. SANDBOX [31] facilitates communication among a group of agents to address social issues thus providing ethically sound data for LLM fine-tuning. Differing from task-oriented agents in the field of recommendation, simulation-oriented agents seek to emulate user behaviors within recommender systems rather than focusing on the recommender. RecAgent [54] attempts to integrate diverse user behaviors in recommendation environments, taking into account external social relationships. Focusing on simulating and evaluating user interactions with the recommenders, our framework Agent4Rec also falls into the category of simulation-oriented agents.

5.2 Recommendation Simulator

Recommendation simulator is a cornerstone in the field of recommendation systems [1, 32, 35, 61]. It diverges from conventional methods that rely on real-world user data, as it excels in replicating or simulating user interactions within recommender systems. This approach offers a cost-effective alternative to online environments and also presents a potential avenue for addressing prevalent challenges such as casual discovery and filter bubbles [4, 18]. Early forays into recommendation simulators primarily serve as rich data sources for subsequent utilization, particularly in the realm of reinforcement learning [20, 46, 50]. For example, Virtual Taobao [50] proposes a virtual environment based on real user interactions to simulate user behaviors on e-commerce platforms. RecSim [20] provides comprehensive toolkits for effectively simulating user behaviors in the setting of sequential recommendation. Also, RecoGym [46] integrates both traditional recommendation algorithms and reinforcement learning framework, followed by evaluation with online and offline metrics. MINDSim [32] simulates user behaviors in a news website. But these traditional simulators all fall short in relatively simple rules, thus lacking flexibility and validity. Recently, LLM-empowered agents have demonstrated significant promise in approximating human-like intelligence, which showcases the considerable potential for recommendation simulators [53, 58]. RecAgent [54] makes the primary attempt to construct a recommendation platform to integrate diverse user behaviors involving movie-watching, chatting, posting, and searching. Instead of behavior integration, we set our sights on in-depth exploration to emulate and evaluate user interactions with both rule-based and algorithm-based recommenders in recommender systems.

6 LIMITATIONS AND FUTURE WORK

Even though Agent4Rec offers a promising research direction in recommender system simulation, we recognize its potential limitations, risks, and challenges that require further exploration and in-depth investigation.

- **Datasource Constraints.** Agent4Rec is implemented exclusively utilizing offline datasets and is primarily constrained by two key factors. First, LLMs necessitate prior knowledge regarding the recommended items, rendering most offline datasets — with only IDs or lacking detailed item descriptions — ill-suited for this task. Furthermore, while online data undoubtedly aligns more naturally with simulators, providing an unbiased perspective to evaluate their effectiveness, acquiring such data poses a considerable challenge.
- **Limited Action Space.** The action space of Agent4Rec is currently limited, omitting critical factors that influence user decisions, such as social networks, advertising, and word-of-mouth marketing. While this simplification facilitates a reliable evaluation of LLM-empowered agents under simple scenarios, it also introduces a gap in real-world user decision-making processes. A key direction for our future work is to encompass a wider spectrum of influential factors to better capture the multifaceted nature of user behaviors, ensuring the simulations are more universally representative of recommendation scenarios.
- **Hallucination in LLM.** Occasional hallucinations have been observed in simulations, such as the LLM failing to accurately simulate human users providing unfavorable ratings to adopted items, fabricating non-existent items and rating them, or not adhering to the required output format. Such inconsistencies can lead to inaccurate simulation outcomes. In light of these observations, our future goal is to fine-tune an LLM specifically for simulating user behavior in recommendation scenarios to enhance the simulator's stability and precision.

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5.1 由 LLM 赋能的生成式智能体

AI 智能体是能够感知其环境、做出决策并采取行动的人工实体 [58]。人类长期以来一直在追求能够处理各种任务的智能体 [8, 21, 41]。近年来，随着大语言模型（LLM）所展现出的卓越能力，大量研究相继涌现 [53]。Generative Agent [42] 是一项开创性工作，设计了配备记忆、规划与反思能力的通用智能体，用以模拟人类的日常生活。在这一通用框架之上，后续的智能体架构可分为面向任务的智能体与面向仿真的智能体 [58]。

任务导向型智能体的核心目标是执行由人类预先定义的任务 [19, 34, 44, 52, 55, 57]。在这一类别中，Voyager [52] 将 LLM 融入 Minecraft 宇宙，使游戏内角色具备高效探索虚拟世界的能力。ChatDev [44] 在软件开发中引入了一种新的智能体协作范式，利用群体智能来简化开发流程。AutoGen [57] 通过定义更复杂的角色并促进多智能体通信，进一步扩展了该范式，以应对多样化的挑战。在推荐系统领域，RecMind [55] 设计了一种由 LLM 驱动的自主推荐智能体，具备自启发式规划能力以及利用外部工具的能力。InteRecAgent [19] 更进一步，通过集成组件增强基于 LLM 的智能体，从而产生既可解释又具备对话能力的推荐器。

面向仿真的智能体旨在在特定场景中复现人类行为，从而实现有价值数据的获取并探索社会议题 [9, 30, 31, 42, 54, 60]。在这些智能体中，[illegible][9] 利用自主智能体来模拟社交平台上公众舆论对热点社会事件的动态演化响应。SANDBOX [31] 促进一组智能体之间的沟通以应对社会议题，从而为 LLM 微调提供符合伦理的数据。不同于推荐领域中面向任务的智能体，面向仿真的智能体试图在推荐系统中模拟用户行为，而非聚焦于推荐器本身。RecAgent [54] 尝试在推荐环境中整合多样化的用户行为，并考虑外部社会关系。我们的框架 Agent4Rec 聚焦于模拟与评估用户与推荐器的交互，也属于面向仿真的智能体类别。

5.2 推荐模拟器

推荐模拟器是推荐系统领域的基石之一 [1, 32, 35, 61]。它不同于依赖真实世界用户数据的传统方法，因为它擅长在推荐系统中复现或模拟用户交互。这种方法为在线环境提供了一种具成本效益的替代方案，同时也为应对诸如随意发现（casual discovery）和过滤气泡（filter bubbles）等普遍挑战提供了潜在途径 [4, 18]。早期对推荐模拟器的探索主要将其作为丰富的数据来源以供后续使用，尤其是在强化学习领域 [20, 46, 50]。例如，Virtual Taobao [50] 提出了一个基于真实用户交互的虚拟环境，用于模拟用户在电商平台上的行为。RecSim [20] 在序列推荐场景下，提供了用于高效模拟用户行为的综合工具包。此外，Reco-Gym [46] 将传统推荐算法与强化学习框架相结合，并随后使用在线与离线指标进行评估。MINDSim [32] 在新闻网站中模拟用户行为。但这些传统模拟器都受限于相对简单的规则，因此缺乏灵活性与有效性。近年来，LLM 赋能的智能体在逼近类人智能方面展现出显著潜力，这也展示了推荐模拟器的巨大潜能 [53, 58]。RecAgent [54] 首次尝试构建一个推荐平台，以整合涉及观影、聊天、发帖与搜索等多样化的用户行为。不同于行为整合，我们将重点放在深入探索上，以在推荐系统中模拟并评估用户与基于规则和基于算法的推荐器之间的交互。

6 局限性与未来工作

尽管 Agent4Rec 为推荐系统仿真提供了一个很有前景的研究方向，我们也认识到其潜在的局限性、风险与挑战，仍需进一步探索并开展深入研究。

- 数据源约束。Agent4Rec 完全基于离线数据集实现，并主要受两个关键因素限制。首先，LLM 需要关于被推荐物品的先验知识，这使得大多数离线数据集——仅包含 ID 或缺乏详细物品描述——并不适合该任务。此外，尽管在线数据无疑与模拟器更为自然地契合，能够提供无偏视角以评估其有效性，但获取此类数据仍然是一个相当大的挑战。

- 有限的动作空间。Agent4Rec 的动作空间目前受到限制，省略了影响用户决策的关键因素，例如社交网络、广告以及口碑营销。尽管这种简化有助于在简单场景下对 LLM 赋能的智能体进行可靠评估，但也在真实世界的用户决策过程中引入了差距。我们未来工作的一个关键方向是纳入更广泛的影响因素谱系，以更好地刻画用户行为的多面性，确保仿真在推荐场景中更具普适代表性。

- 大语言模型（LLM）中的幻觉。在仿真中观察到偶发的幻觉现象，例如：LLM 无法准确模拟人类用户对已采纳物品给出不利评分、捏造不存在的物品并对其评分，或不遵循所要求的输出格式。这类不一致性可能导致不准确的仿真结果。鉴于这些观察，我们未来的目标是专门为推荐场景中的用户行为仿真对 LLM 进行微调，以提升模拟器的稳定性与精确性。

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